

Chapter 13

System Operation and Maintenance

SDLC

Brief Introduction to SDLC

Phase 1

System Request

System Planning

Phase 2

System Analysis

Preliminary Investigation Report

Phase 3

System Design

System Requirements Specification

Phase 4

System Development

System Design Specification

Phase 5

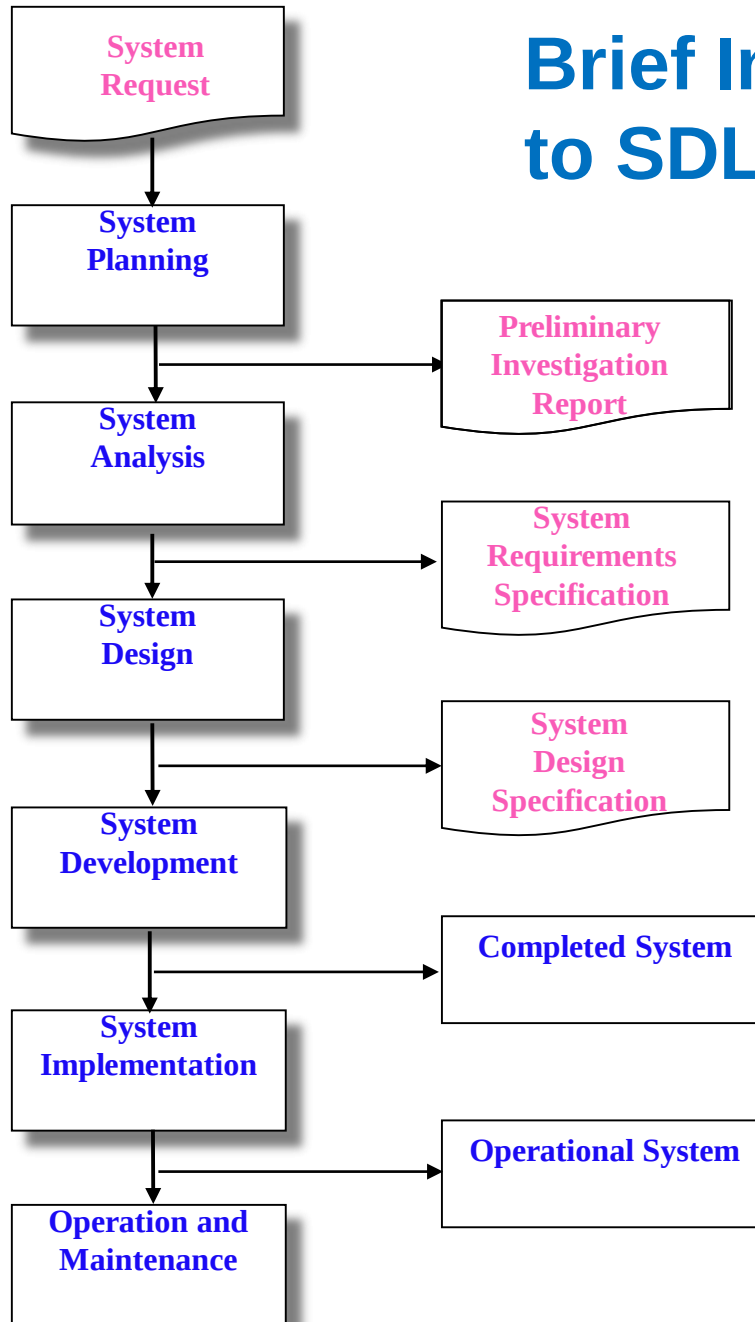
System Implementation

Completed System

Phase 6

Operation and Maintenance

Operational System



System Operation

- **Meaning**

- System in operation until it cannot provide any useful functions to the users anymore.

- **Purpose**

- Support Users in using the system to help them perform their works.

- **Sources of support**

- In-house (your company IT staffs develop and support the system) and Vendors (your company buys system from vendor)

- **Forms of support**

- User training, Help desks, Online and Automated support

Forms of Operation Supports

1. User Training

- Teach **users** how to use the system
- Support users when they face problems using the system.
- System undergone changes or enhancement
 - If changes are **major** – use formal training
 - If changes are **minor** – use e-mail support, training manual supplements, user guides, online help facilities and special training Web site.

2. Help Desks

- Help the users in various ways :
 - **Answer** operational or technical questions.
 - Teach users how to **use** the system to meet their information needs
 - Teach users how to use system resources in more **effective** manner
- Differentiated from Maintenance
 - **Help desk** - provide **assistance** to the users to be more effective in performance of their jobs and lead the users solving problems for themselves
 - **Maintenance** - aims to provide continuous and uninterrupted **supply** of computer resources to the users.
- Help desks **personnel**. Strong technical and interpersonal skills, solid understanding of the business, well trained for the new system.

3. Automated Supports

- **Purpose** : more supports moved online and automated to cut down manual cost of providing support.
- **Examples** : e-mail responses, FAQs, bulletin boards, automated voice mail, and online help facilities.
- **On-line support forums**. Allow users access to information on bugs, their new product releases, and tips for more effective usage of system. Online services over company intranets.

3. Automated Supports

- **Online chat rooms.** Eg. Blackboard, popularly used in universities and colleges to help deliver online courses and to allow students and lecturers to meet online.
- **Online knowledge.** Technical and support information about vendor products and information for on-site personnel to use in solving problems. Some vendors allow users to access some of these information.
- **Voice-response systems.** Allow users to search for some pre-recorded messages on various issues, such as, usage, problems and solutions. These messages are also called podcasts.

Examples of Operation Support

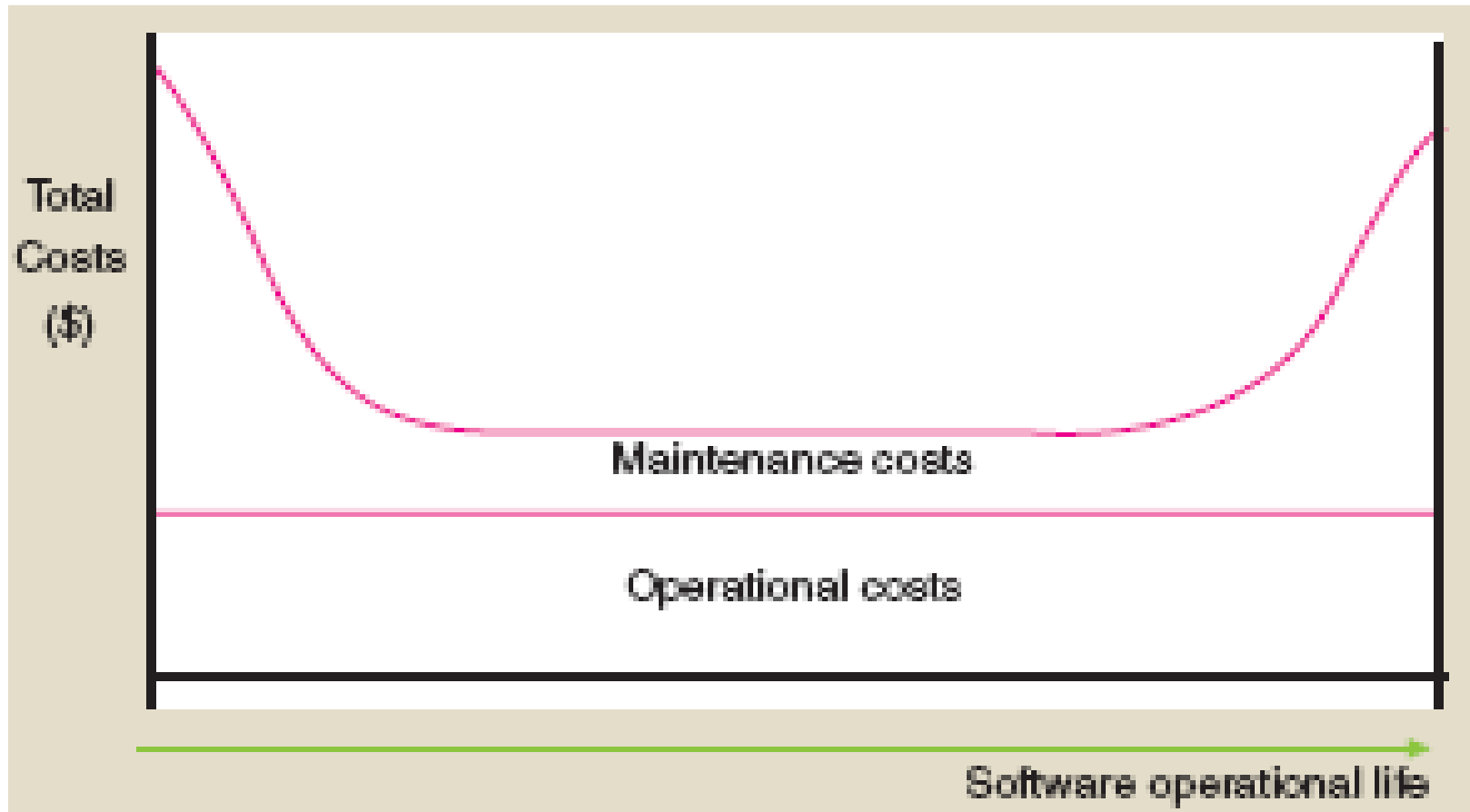
1. Assist users in **writing** simple programs in end-user computing
2. **Install** new hardware or software for the users
3. Assist users in **evaluating** a particular software
4. Help users to submit system **proposal**
5. **Teach** users (new and old) how to use the system
6. Help users to solve **problems** encountered while using the system
7. Assist users in **extract** data from server databases onto their personal computers
8. Set up new user **accounts** for a particular system

System Maintenance

What is Maintenance ?

- **Operational life**
 - **Begins** : System is implemented
 - **Ends** : System no longer useful eg. not economical, replaced etc
- **Operation Vs Maintenance**
 - System maintenance → IT Personnel
 - System operation → End-users
- **Purposes** : to ensure that the system will **continue** to be functioning at an **acceptable** level to the users by improving or modifying the system.

Costs of Maintenance



- **Diagram shows :**
 - **Operational cost** – quite constant
 - **Maintenance cost** – varies in the shape of 'bath-tub' curve

3 Types / Purposes of Maintenance

1. Perfective (Users Requests)

- **Meaning :**

- To cope with **new** requirements due to **users' requests** (eg. new markets, new information required, new technologies)
- The requests by users may include to improve software **functionality, usability** and **performance**.

- **Examples**

- Improving **performance** eg. faster **response** to customer service as requested by users.
- Improving **functionality** eg to add new features or **reports** as requested by users.
- Improving **usability** eg improving **layouts** of some reports in an inventory system as requested by users.
- Improving **usability** eg improving the **interface** (to make the system easier to use).

2. Corrective (Correction of Errors)

- **Meaning**
 - To **fix bugs** or faults (in programs) which are unforeseen and arise during operation.
 - To keep the programs working **correctly**.
- **Examples**
 - Diagnose and fix logic errors or program code.
 - Update software drivers
 - Restore proper configuration settings

3. Adaptive (Changes in Environment)

- Meaning

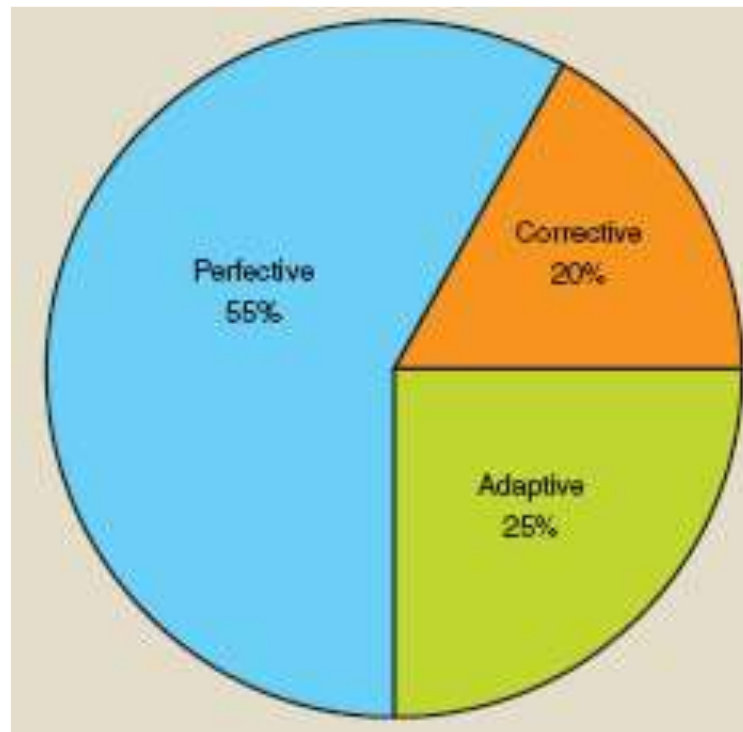
- To **adapt** the system due to changing environment :
 - business environment (eg. new regulatory requirements)
 - computer environment (eg. hardware or software changes such as a change of OS)

- Examples

- Changes to payroll system due to new employment *laws*.
- Add *new online capability* due to business environment changes
- Create *new reports* due to business environment changes
- Add *new data entry field* to input screen due to business environment changes
- *New systems* will be added eg. OS, hardware, DBMS

Distribution Costs Of Maintenance

- Maintenance costs is significantly very high up to up **70%** of the total life cycle costs.
- Percentage distribution of maintenance



Reducing Maintenance Costs

- **Thorough Analysis of Requirements.** If systems can be correctly and fully specified and agreed there will be little need for maintenance.
- **Predictive Design.** Any likely changes in requirements could be built into the software at design stage thus eliminating or reducing the need for maintenance.
- **Decomposition of Program** (modularity). This can help maintenance. It is easier to make changes to small programs than larger ones.

Reducing Maintenance Costs

- **Thorough Program Testing.** This would reduce the incidence of errors in the programs during operation.
- **Complete Documentation.** Up-to-date documentation would permit changes to be made easily and efficiently.
- **Usage of Methodology and Productive Tools.** Efficient methodologies, such as prototyping and structured approach and use of CASE tools can help to carry out maintenance.

Exercise for Students

- **Identify an appropriate type of maintenance which each of the following can reduce :**
 - Conduct thorough program testing
 - Carry out predictive design taking into accounting potential features required.
 - Keep complete and up-to-date documentation for the system.
 - Design the system to be more modular.
 - Conduct thorough analysis of requirements.
 - Use efficient methodology and productive tools.

Suggested Answers

- Conduct thorough program testing. (**perfective**)
- Carry out predictive design taking into accounting potential features required. (**perfective**)
- Keep complete and up-to-date documentation for the system. (**all types**)
- Design the system to be more modular. (**all types**)
- Conduct thorough analysis of requirements. (**perfective**)
- Use efficient methodology and productive tools. (**all types**)

Software Maintenance Contract

- **Meaning**

- **Parties** : Between the customer and the software supplier
- **Fees** : Customer pays an annual fee (usually some percentage of the original software cost).
- **Time** : Specific timeframe and renewable.

- **Content or Services**

- **Information**. Extra information (eg. Factsheets, newsletter) about using the package
- **Help**. Helpline facility allows users to resolve problems
- **Updates**. Fixing software faults emerged.
- **Upgrades**. New version at discounted price.
- **Legal conditions** eg. Duration, customers' obligations to use software. Liability of supplier.

User Groups

- A **user group** is a forum for users of particular hardware or, more usually, software, so that they can share ideas and experience and, on occasions, acting as an arbiter (**intermediary**) in disputes with supplier.
- Ways to Set up
 - **By Software Manufacturers.** Software manufacturer use them to maintain contact with customers and as a source of new product ideas.
 - **By Groups of Users.** Users who were not satisfied with the level of support they were getting from suppliers of proprietary software.

User Groups

- **Benefits to Users**
 - **Share ideas.** Users of a particular package can meet, or perhaps exchange views over the Internet to discuss solutions, ideas or 'short cuts' to improve productivity.
 - **Feedbacks.** Ideas from members and IT specialists can be incorporated into future system development.
 - **Arbiter.** Acts as an arbiter in any disputes between users and suppliers eg functionality disputes.

Examples of User Groups

- **Hardware platforms**
 - 3Com Users Group (3CUG)
 - IBM mainframe - SHARE
 - Boston Mac Users Group
- **Software platforms**
 - Adobe User Group - Philippines
 - International Informix Users Group (IIUG)
 - UK Oracle User Group
 - ASUG (Americas' SAP Users' Group)

Example

Oracle User Group Community

A worldwide community of
independent user groups.



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